



LEARN
PROMPT
DISCOVER
PRACTICE

Claude 101

Foundations, Prompting, Files, and Everyday Use

LEARN / PROMPT / EXPLORE / CREATE / APPLY



Foundations

START
HERE



Models

KNOW
THE OPTIONS



Prompting

ASK
BETTER



Files

WORK WITH
DOCUMENTS



Use Cases

WRITE, RESEARCH,
ANALYZE & CODE

START
WITH
CONFIDENCE

IDEAS
INTO
ACTION

YOUR CLAUDE JOURNEY STARTS HERE

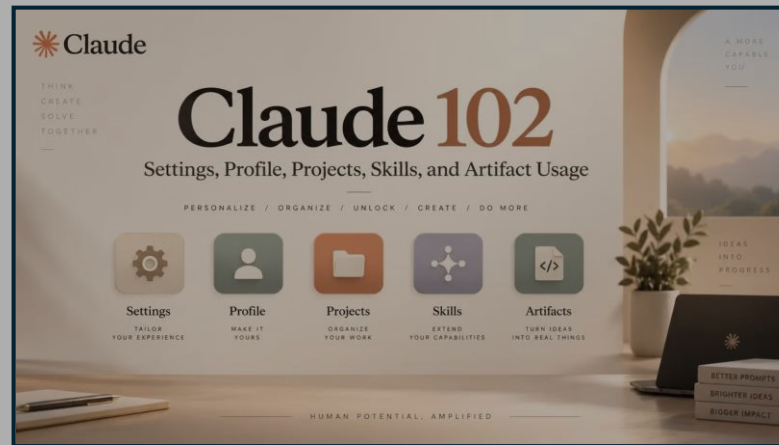
BETTER PROMPTS
SMARTER WORK
REAL RESULTS

Three sessions, one journey



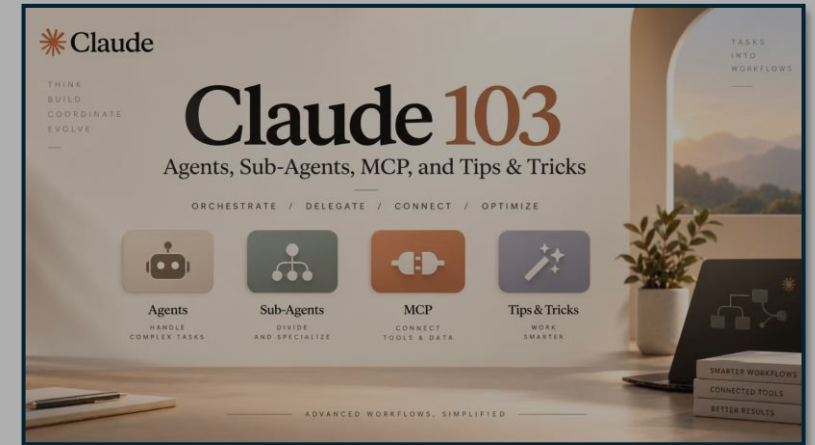
101 • Foundations

What Claude is and where it fits
Model families
Prompting and context
Documents and files



102 • Workspace

Settings, profile, and preferences
Projects: files, instructions, context
Skills: reusable workflows
Artifacts: docs, code, diagrams



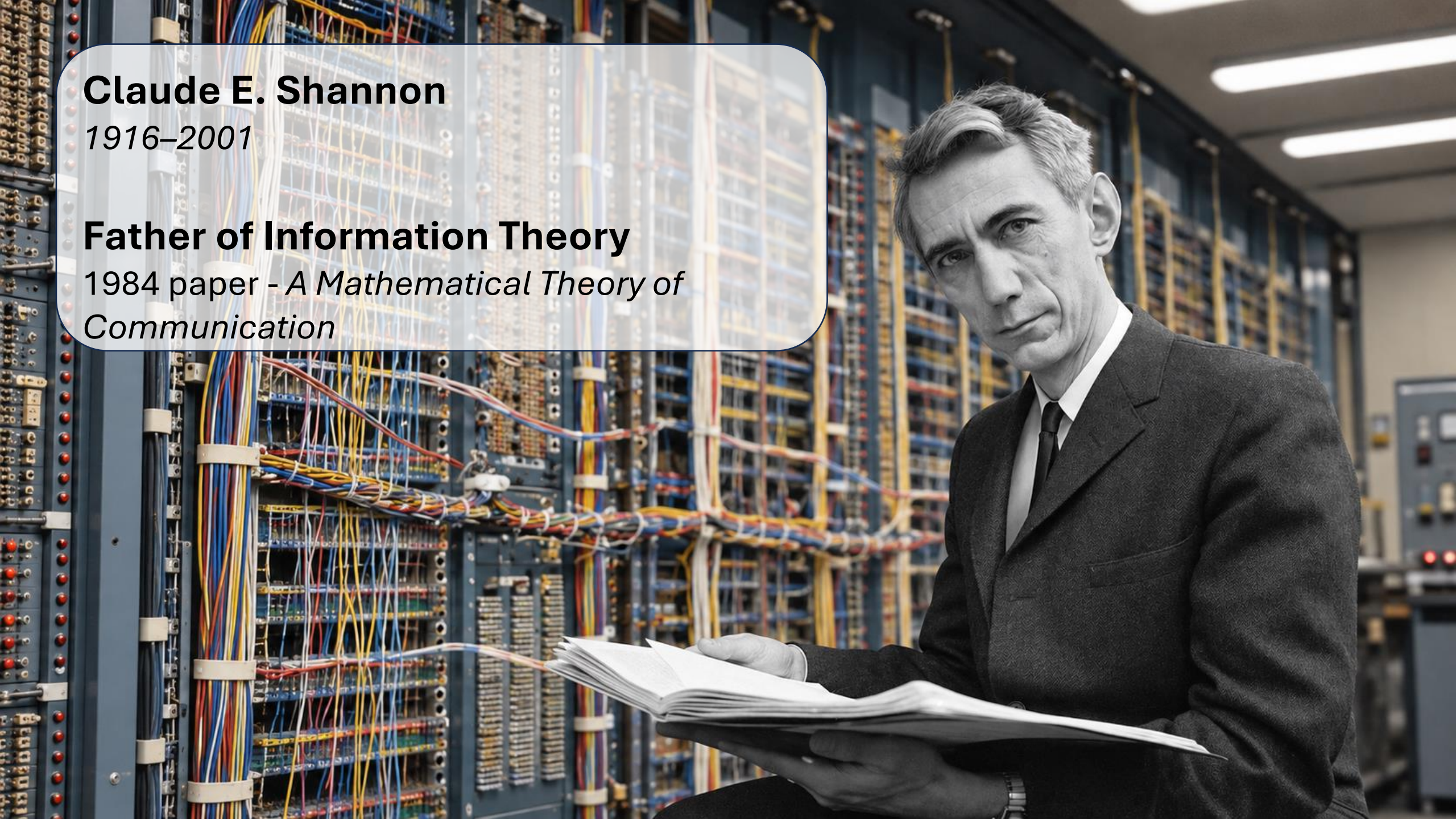
103 • Advanced

Agents + sub-agents
Coordinating and delegating work
Model Context Protocol (MCP)
Automation + integrations

Agenda

- What is Claude?
- Subscription Tiers
- Models
- Prompting
- Documents & Files
- Best Practices
- Demos throughout the presentation
- Q&A



A black and white photograph of Claude E. Shannon, a pioneer in information theory, standing in a large room filled with rows of telephone exchange equipment. He is wearing a dark suit and tie, and is holding a large, open book or document. The background is filled with complex wiring and mechanical components, typical of early 20th-century telecommunications infrastructure.

Claude E. Shannon

1916–2001

Father of Information Theory

1984 paper - *A Mathematical Theory of Communication*



Claude is an artificial intelligence, trained by Anthropic using Constitutional AI to be safe, accurate, and secure — the trusted assistant for you to do your best work.

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

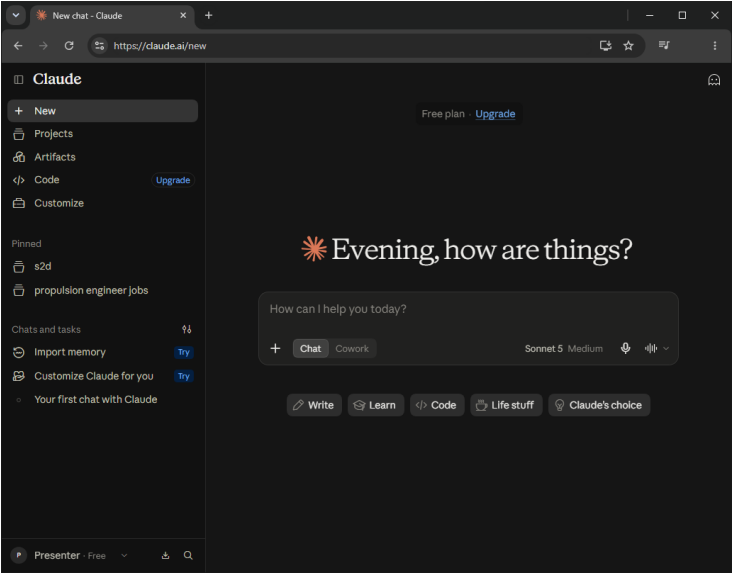
Claude Security

Features

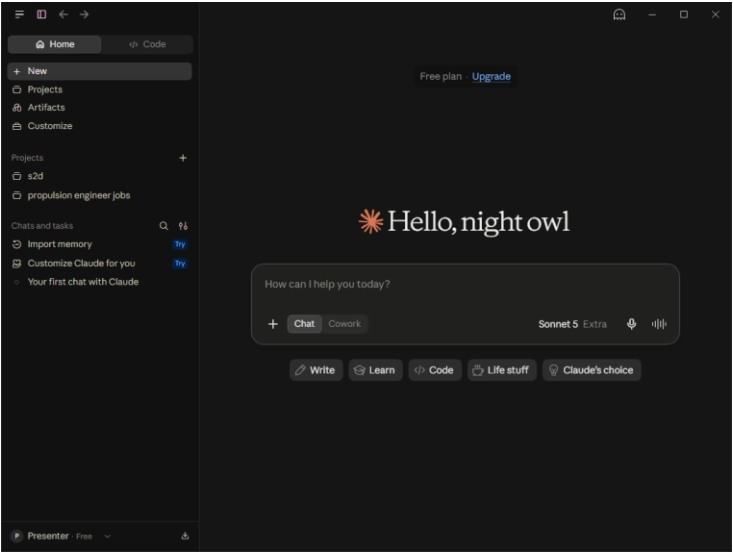
Claude in Chrome

Claude for Slack

Claude for Microsoft 365



Web



Desktop



Phone

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365

Claude Code

PS C:\Projects\azure-ml-demo> claude

Claude Code

Welcome to Claude Code!
Sonnet 4.6 • Anthropic
Type /help for available commands.

Tips

/help

Show all commands

/docs

View documentation

/context

See what's in the context window

/compact

Reduce context usage

/clear

Start a fresh conversation

> /context

Context Window Usage

Model: Claude Sonnet 4.6 Context Window: 1,000,000 tokens

42,380 / 1,000,000 tokens (4.2%)

Breakdown by Source

Conversation history

18,240 tokens (1.8%)

Project instructions

8,120 tokens (0.8%)

Files (3)

12,450 tokens (1.2%)

Tools / tool results

2,310 tokens (0.2%)

System / other

1,260 tokens (0.1%)

Loaded Files (3)

README.md

4,210 tokens

src/main.py

5,840 tokens

data/sample.csv

2,400 tokens

Project Instructions

CLAUDE.md

8,120 tokens

Recent Messages (last 5)

User

Explain how this script works

320 tokens

Assistant

The script loads the CSV data and...

1,120 tokens

User

Add error handling

210 tokens

Assistant

Here's an updated version with...

980 tokens

User

/context

6 tokens

Tip: Use /compact to reduce context usage or /clear to start a fresh conversation.

>

Claude Code v1.0.0 • Anthropic

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365

Anthropic's organizational tier and agentic coding tool designed to let engineering teams run Claude's coding assistant natively in terminals and IDEs with centralized security, governance, and administration.

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

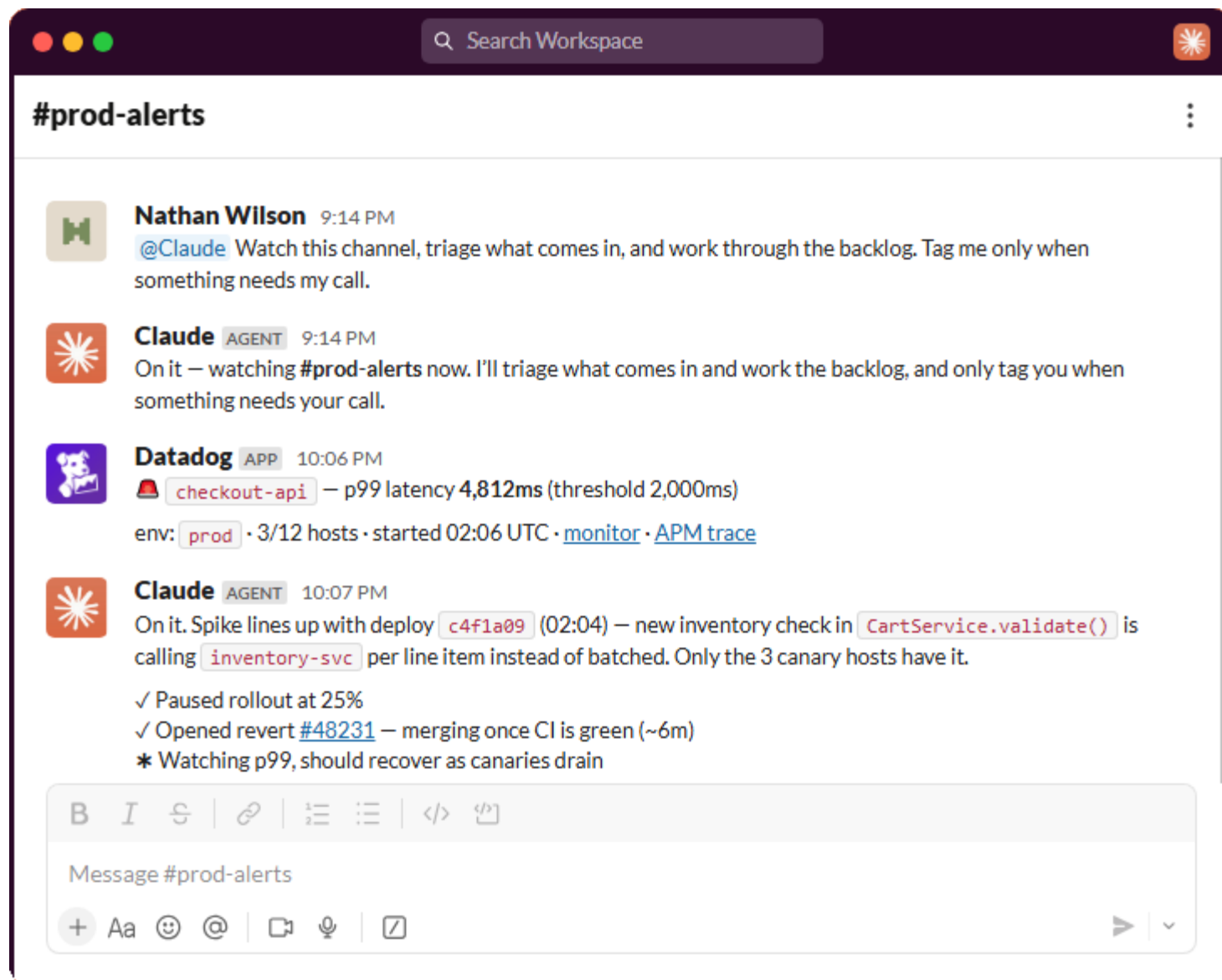
Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365



Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365

AI-powered visual creation tool by Anthropic Labs that generates polished designs, interactive prototypes, slide decks, and marketing assets directly from text prompts.

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365

Claude Science

Beta

< scRNA-seq ▾

+ New

Customize

Files

Active

◦ SCVI Hyperparameter ... 8

SCVI Hyperparameter Screen

Dispatching the 8-arm scVI sweep to lab_cluster A100s — $n_{\text{latent}} \in \{10, 20, 30, 50\} \times n_{\text{layers}} \in \{1, 2\}$, 40k cells $\times$ 2,000 HVGs, batch_key="sample_id", 50 epochs, seed 0.

arm	n_{latent}	n_{layers}	label
1	10	1	d=10 L=1 · scVI COVID-PBMC (1/8)
2	10	2	d=10 L=2 · scVI COVID-PBMC (2/8)
3	20	1	d=20 L=1 · scVI COVID-PBMC (3/8)
4	20	2	d=20 L=2 · scVI COVID-PBMC (4/8)
5	30	1	d=30 L=1 · scVI COVID-PBMC (5/8)
6	30	2	d=30 L=2 · scVI COVID-PBMC (6/8)
7	50	1	d=50 L=1 · scVI COVID-PBMC (7/8)
8	50	2	d=50 L=2 · scVI COVID-PBMC (8/8)

Stephenson 2021 COVID PBMC — 40k cells, 2,900 batch-aware HVGs, no integration

By sequencing site

By author cell type

REMOTE: 8

- lab_cluster · d=10 L=1 · scVI CO... 16m 2s
- lab_cluster · d=10 L=2 · scVI ... 15m 42s
- lab_cluster · d=20 L=1 · scVI C... 15m 19s
- lab_cluster · d=20 L=2 · scVI ... 14m 58s
- lab_cluster · d=30 L=1 · scVI C... 14m 36s
- lab_cluster · d=30 L=2 · scVI ... 14m 16s

8 running - 16m 2s

Ask anything

+  

Notebook

pbmc-pipeline Shared with the agent Live ▾

[28] python

```
1
2 import pandas as pd
3 pd.set_option('mode.string_storage', 'python')
4 import numpy as np, scanpy as sc, anndata as ad, scipy.sparse as sp
5
6 a = sc.read_h5ad("covid_pbmc_40k_hvg.h5ad")
7 print("loaded:", a.shape, "uns keys:", list(a.uns.keys()))
8
9 # minimal, version-portable object: counts + obs + var only
10 keep_obs = ["sample_id", "donor_id", "Site", "initial_clusterin",
11             "author_cell_type", "disease", "Status",
12             "Status_on_day_collection_summary", "cell_type"]
13 keep_obs = [c for c in keep_obs if c in a.obs.columns]
14 obs = a.obs[keep_obs].copy()
15 for c in obs.columns:
16     obs[c] = pd.Index(np.asarray(obs[c], dtype=object))
17 obs.index = pd.Index(np.asarray(obs.index, dtype=object))
18 var = pd.DataFrame(index=pd.Index(np.asarray(a.var_names, dtype=object),
19
20 clean = ad.AnnData(
21     X=sp.csr_matrix(a.layers["counts"]),
22     obs=obs, var=var,
23 )
24 clean.layers["counts"] = clean.X.copy()
25 clean.write_h5ad("covid_pbmc_40k_hvg.h5ad")
26 print(f"rewritten: {os.path.getsize('covid_pbmc_40k_hvg.h5ad')}/
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```

output

wrote covid_pbmc_40k_hvg.h5ad

Python — pbmc-pipeline kernel

Connected to the agent's live kernel — variables and state are shared. Type an expression and press Enter.

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365

AI-powered tool built into Claude that scans software codebases for security flaws and suggests targeted patches for human review.

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365

Official browser extension by Anthropic that allows the Claude AI assistant to view and interact directly with your Google Chrome browser.

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

Claude in Chrome

Claude for Slack

Claude for Microsoft 365

AI integration that brings Anthropic's Claude directly into your Slack workspace as a collaborative team member.

Products

Claude

Claude Code

Claude Code for Enterprise

@Claude

Claude Design

Claude Science

Claude Security

Features

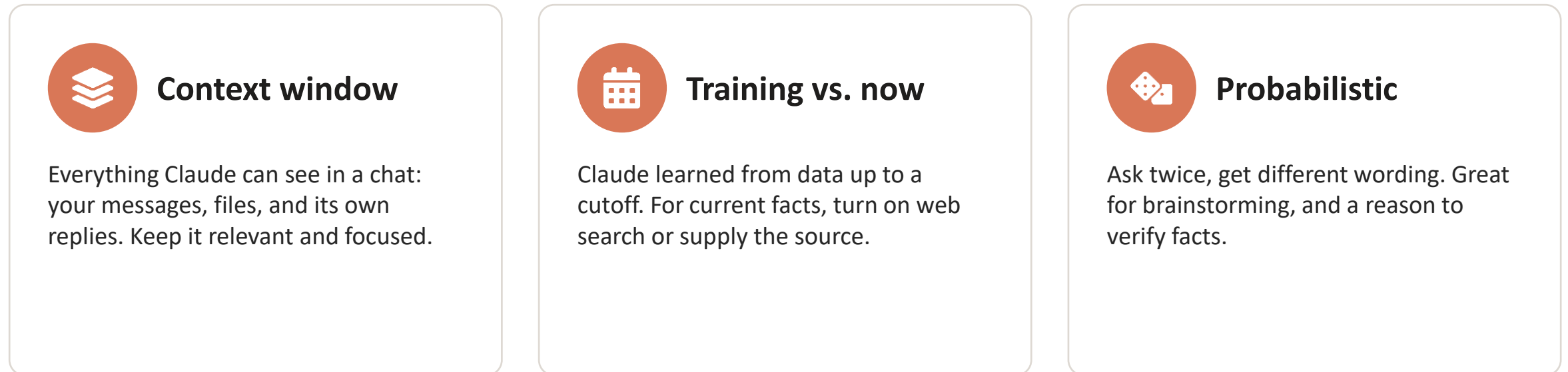
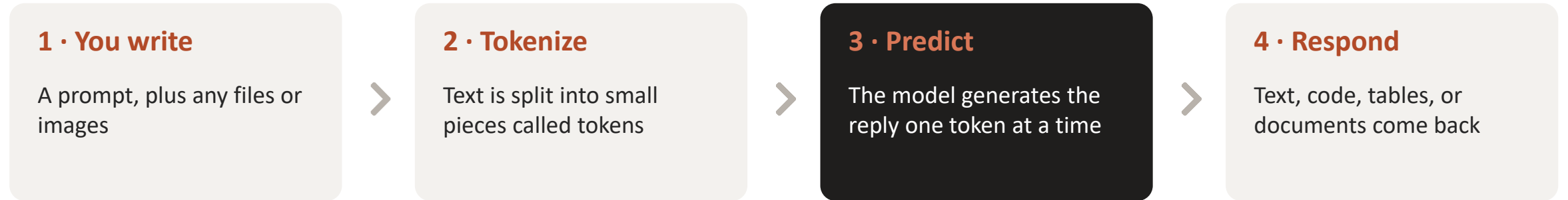
Claude in Chrome

Claude for Slack

Claude for Microsoft 365

An AI integration that brings Anthropic's Claude assistant directly into productivity apps like Word, Excel, PowerPoint, and Outlook.

How it works, in 60 seconds



Pick the model by risk and complexity—not habit

Start with the cheapest/fastest model that reliably meets your quality bar.

Task	Likely starting point	Why
Extract fields from 500 tickets	Haiku	Quick answers, summaries, and simple extraction — anything you want done instantly
Refactor a service with tests	Sonnet	Coding, writing, analysis, and multi-step workflows — your versatile default
Architecture decision with competing constraints	Opus	Deep research and complex reasoning that genuinely needs sustained thinking
Long-horizon agentic research	Fable	Your largest, most critical projects: long, complex tasks Claude works through with fewer check-ins
Architecture decision with competing constraints	Mythos	Only available through Cybersecurity and Life Sciences Verification Program

Production rule: define success criteria and test representative prompts before standardizing on a model.

Tiers

Plan	Price	Claude Code?	Best for
Free	\$0	No	Occasional/introductory use
Pro	\$20/mo	Yes	Regular individual use
Max 5x	\$100/mo	Yes	Heavy individual use
Max 20x	\$200/mo	Yes	Very heavy/ daily users
Team Standard	\$25/seat/mo	No	Teams 2 -150
Team Premium	\$125/seat/mo	Yes	Team power users
Enterprise	Seat price + usage at API rates	Yes	Larger orgs, compliance
API pay-as-you-go	Per-token	Yes	Spiky usage, cost-conscious teams

Model costs

				
	<u>Claude Fable 5.1</u>	<u>Claude Opus 5.5</u>	<u>Claude Sonnet 5</u>	<u>Claude Haiku 4.5</u>
	For demanding reasoning and long-horizon agentic work	For long-running agentic coding and knowledge work	The best combination of speed and intelligence	The fastest model with near-frontier intelligence
Comparative latency ⓘ	Slower	Moderate	Fast	Fastest
Pricing ⓘ	\$10 / input MTok \$50 / output MTok	\$4 / input MTok \$20 / output MTok	\$2 / input MTok \$10 / output MTok	\$1 / input MTok \$5 / output MTok
Capabilities				
Thinking ⓘ	Adaptive (always on)	Adaptive (always on)	Adaptive	Extended
Default effort ⓘ	high	medium	high	Not supported
Context window ⓘ	1M tokens	1M tokens	1M tokens	200K tokens
Max output ⓘ	128K tokens	128K tokens	128K tokens	64K tokens
Reliable knowledge cutoff ⓘ	Jun 2026	Jun 2026	Jan 2026	Feb 2025

* Save 50% with batch processing.

<https://docs.claude.com/en/docs/build-with-claude/batch-processing>

<https://platform.claude.com/docs/en/models/overview>

Demo



Print StartTime

Extract all dates and format them as YYYY-MM-DD in a bulleted list: 'The project kicked off on March 5th, 2024. We hit our first milestone on July 12 of the same year, and final delivery happened on Jan 15, 2025.'

When done after outputting response Print EndTime and Print ElapsedTime

Exploring Model Differences

Use new session for each run across Haiku, Sonnet, Opus, and Fable.

Meet Claude

Claude is an AI assistant from Anthropic, designed to help with writing, research, analysis, coding, and everyday problem-solving.

You bring text, files, and images. Claude responds with text, code, tables, and documents.

Built by Anthropic, an AI safety company, with a focus on being helpful, honest, and harmless.

Available in chat apps, developer tools, and an API.



Writing

Draft, edit, and polish



Research

Explore and compare



Analysis

Make sense of data



Coding

Write, explain, debug

Strengths and limits



Where Claude shines

- Drafting and editing emails, docs, and summaries
- Summarizing and extracting from long documents
- Explaining unfamiliar concepts, code, and error logs
- Brainstorming options and structuring ideas
- Reformatting: notes to email, CSV to table, code to another language



Where to be careful

- Can sound confident and be wrong: verify facts, numbers, and code
- Doesn't know recent events unless it searches the web
- Exact math over big data: ask it to show work or run code
- Doesn't know your organization unless you tell it
- Sensitive data: follow your company's AI and privacy policy

Claude is a reasoning interface—not a search box

The quality of the result depends on the task definition, context, and evidence you provide.

- Use natural language to assign a task, role, constraints, and desired output.
- Treat responses as generated work products that require validation, not guaranteed truth.
- Add source material when the answer depends on your environment, documents, or current facts.
- Iterate conversationally: critique, refine, compare, and ask Claude to expose assumptions.

Intent

What outcome do I need?

Context

What does Claude need to know?

Reasoning

What trade-offs matter?

Output

What format is useful?

Verify

What must I check?

A strong prompt has five parts

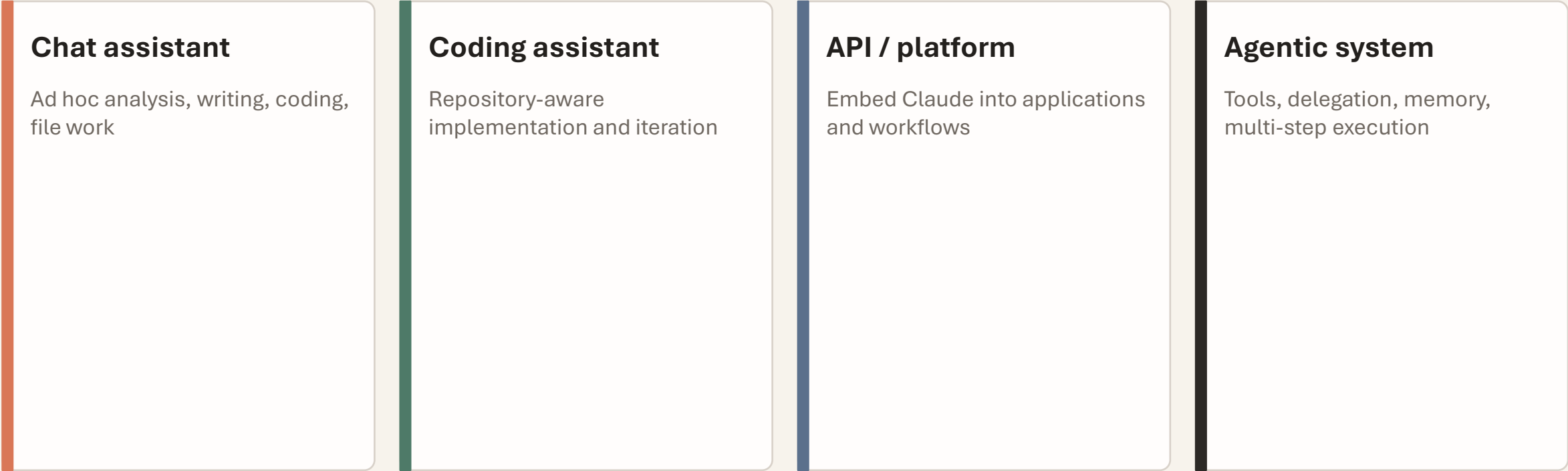
You do not need a magic phrase—you need an unambiguous task contract.

- 1 ROLE / POV** “Act as a senior cloud architect...”
- 2 TASK** “Review this deployment plan...”
- 3 CONTEXT** “We run .NET 8 on AKS...”
- 4 CONSTRAINTS** “No downtime; budget +15% max...”
- 5 OUTPUT** “Return risks, mitigations, and a decision table.”

Optional sixth part: examples. A good example often beats another paragraph of explanation.

Where Claude fits in the generative AI landscape

Use the right interface for the work: chat, code assistant, API, or agentic system.



Claude 101 lives mostly here ↑

102 expands the workspace • 103 expands the system

From vague request to useful engineering prompt

The biggest quality jump usually comes from adding context and a definition of “good.”

Vague

“Review this architecture and tell me what you think.”

Problems

- No goal or audience
- No constraints
- No decision criteria
- No required output shape

Specific

“You are reviewing a .NET 8 API on Azure. Identify the top 5 reliability risks before production. Assume 99.9% SLO, single-region deployment, Azure SQL, and a 15-minute RTO. Return: risk, evidence, impact, mitigation, and a question to validate.”

Context is the highest-leverage prompt ingredient

PROMPTING

Give Claude the facts it cannot safely infer.

Environment

Versions, stack, architecture,
constraints

Audience

Developer, executive,
customer, operator

Evidence

Logs, files, snippets,
requirements, source links

Boundaries

What not to change;
security/compliance rules

Beginner rule: when Claude “guesses,” ask yourself what context you expected it to know but never provided.

Structure the prompt when the task gets dense

PROMPTING

Headings, delimiters, and examples make complex requests easier to follow and revise.

```
<task>
Review the incident summary and recommend next steps.
</task>

<context>
- .NET 8 API on AKS
- p95 latency jumped after 14:00 UTC
- CPU remained normal
</context>

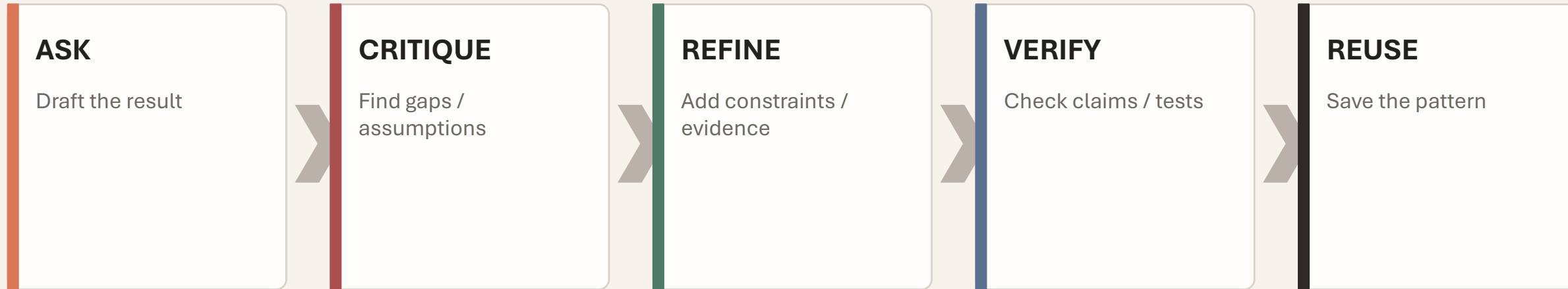
<output>
1. Likely causes ranked by evidence
2. Checks to run next
3. What would falsify each hypothesis
</output>
```

- Use semantic headings or XML-style tags for long prompts.
- Separate evidence from instructions.
- Ask for falsification criteria when diagnosing.
- Use examples when format consistency matters.

Prompting is an iteration loop, not a one-shot contest

PROMPTING

Use Claude to improve the task definition as well as the answer.



Useful follow-up: “Before you answer again, list the assumptions in your previous response that could materially change the recommendation.”

DEMO 1 • Turn a weak prompt into a technical copilot

LIVE DEMO

Scenario: triage a production incident from messy notes.

Step 1 • Baseline

Paste the incident notes and ask:

“Why is this app slow?”

Step 2 • Add context

Specify stack, SLO, time window, known-good signals, and what changed.

Step 3 • Force rigor

Ask for ranked hypotheses, evidence, next checks, and falsification criteria.

Goal: show the audience that better results come from better task specification—not “secret prompt words.”

Six techniques you can use today

1

Let Claude interview you

Before you answer, ask me up to 5 clarifying questions.

2

Show, don't just tell

Here's a memo whose tone I like. Match it:
[paste]

3

Ask for a plan first

Outline your approach in 5 bullets. Wait for my OK.

4

Iterate in the same chat

Shorter. More formal. Add a risks section.

5

Ask for options

Give me 3 approaches with trade-offs, then recommend one.

6

Have Claude check its work

What assumptions did you make? What could be wrong here?

Files turn Claude from “general assistant” into “your-context as

FILES

Attach the evidence instead of paraphrasing everything into the prompt.

- Upload documents, data, images, and source material directly into a chat.
- Ask Claude to summarize, compare, extract, transform, critique, or reason across files.
- Reference page numbers, columns, filenames, and sections in your instructions.
- For important decisions, require citations back to the supplied evidence.

PDF

reports • diagrams

DOCX / TXT

requirements • notes

CSV / XLSX

tabular analysis

JSON / HTML

structured data

Images

screenshots • diagrams

Know the file-processing boundaries

Limits change over time, so verify them before a production workflow depends on them.

500 MB max per chat-uploaded file	20 files max files per chat	1000 pages maximum PDF length	≤100 pages PDF text + visual analysis	101–1000 PDF text only
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Non-PDF documents are primarily text-extracted. Embedded images in those files may not be interpreted the same way as a multimodal PDF.

Source: Claude Help Center, “Upload files to Claude,” updated July 23, 2026.

Ask file questions that produce traceable outputs

FILES

Good file prompts point Claude to evidence and force useful structure.

Summarize

“Give me a 10-line executive summary and cite the pages behind each major claim.”

Compare

“Compare requirements-v1.docx vs v2.docx. List additions, removals, and changed acceptance criteria.”

Extract

“From the CSV, group cost by resource group and identify the top 5 contributors.”

Critique

“Find contradictions between the architecture diagram and the written deployment steps.”

DEMO 2 • Analyze multiple files like an operator

LIVE DEMO

Scenario: combine cost data with a change log to find where to investigate first.

Files

cloud-costs.csv
change-log.txt

Ask Claude to use both—not just summarize them independently.

Prompt

“Find the top cost movers and correlate them with recent changes. Separate facts from hypotheses.”

Verify

Require file/row references, confidence, and questions that would confirm or reject the suspected cause.

Trust but verify common failure modes

BEST PRACTICES

Fluent output can still be incomplete, outdated, or wrong.

Hallucination

A plausible detail with no evidence

Hidden assumptions

Claude fills gaps you never specified

Stale facts

Current-state questions depend on fresh sources

Overconfidence

Weak evidence presented too strongly

Countermeasures: provide sources • ask for uncertainty • demand citations • run tests • independently verify high-impact claims

Claude 101 – Key Takeaways

- If new start with the free Tier
- Learn which models/effort are most effective for your tasks
 - Start with cheaper/faster models
- Concentrate on writing better prompts
- Validate important work





References

Daniel Colón LinkedIn

<https://www.linkedin.com/in/danielecolon/>

Presentation Slides/Code

<https://github.com/danielecolon/Claude>

Get started with Claude

<https://support.claude.com/en/collections/9811058-get-started-with-claude>